

Emotional Models: Supervised vs. Zero-Shot Classification

Miguel Castela¹ and Miguel Martins²

Univsersity of Coimbra, Portugal
uc2022212972@student.uc.pt
uc2022213951@student.uc.pt

1 Introduction

This project explores two distinct approaches to emotion classification in text using Transformer-based models: supervised learning with feature extraction and zero-shot classification via Natural Language Inference (NLI).

2 Dataset Description

The dataset utilized in this project is the `dair-ai/emotion` dataset. It consists of text samples, primarily collected from Twitter, labeled with one of six emotions: *joy*, *sadness*, *anger*, *fear*, *love* and *surprise*. The dataset is split into training, validation, and test sets, with the training set containing 16,000 samples, the validation set 2,000 samples, and the test set 2,000 samples. The samples are distributed according to the following table:

Table 1. Class distribution of the `dair-ai/emotion` dataset.

Emotion	Train	Validation	Test	Total (%)
Joy	5,362	704	695	33.8%
Sadness	4,666	550	581	29.0%
Anger	2,159	275	275	13.5%
Fear	1,937	212	224	11.9%
Love	1,304	178	159	8.2%
Surprise	572	81	66	3.6%
Total	16,000	2,000	2,000	100.0%

2.1 Foreseeable issues

Class imbalance By looking at table 1, we can see that the dataset is imbalanced, with the *joy* and *sadness* classes being significantly more represented, together accounting for over 60% of the data, and *surprise* being the least represented by a considerable margin, 3.6%. This imbalance could lead to biased model performance, favoring the majority classes, and already gives away that “Accuracy” might not be the best metric to evaluate the models.

Overfitting to keywords Another issue present is the overfitting of words or terms to a certain emotion. For example, the word “happy” is strongly associated with the *joy* class, and its presence in a text sample could lead the model to predict *joy* regardless of the overall context. One example of this is [por exemplo...].

3 Methodology

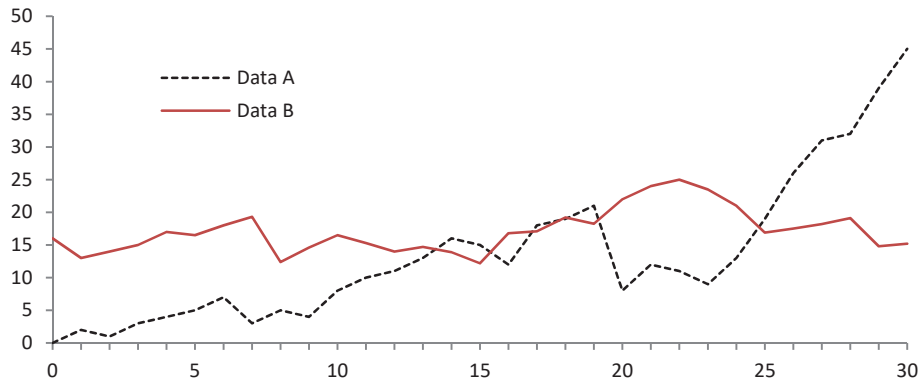


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

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Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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